

Reading and Making Line Plots

Answer Key

Grade 2–3

Quick Answers

1. 3	4. 6	7. 12	10. 0
2. 3	5. 3	8. 2	
3. 16	6. 4	9. 5	

Curriculum

Level: Grade 2–3

2.MD / 3.MD – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 11 tagged checks.

Common wrong answers

5. *If they got 9: added the two stacks of X's instead of finding how many more one has than the other*

7. *If they got 6: counted the marks on the number line instead of counting the X's*

8. *If they got 1: subtracted the two lengths written under the line instead of the numbers of X's above them*

9. *If they got 6: counted every mark on the number line from the shortest to the longest instead of subtracting*

What this sheet is teaching, and what to listen for. A line plot holds two different kinds of number, and almost every error on this sheet is the two of them swapped. The numbers *along the bottom* are the thing being counted — books, or centimetres. The X's *above* them are how many times that thing happened. When a child gives you a wrong answer, ask “were you counting books or were you counting children?” rather than pointing at the plot.

1. Count one stack. How many children read exactly 4 books?

Put a finger on the 4 at the bottom of the plot and count straight up. There are three X's in that stack, and each X is one child. 3

Say what the answer counts. The answer is three *children*, not three books. A child who answers “4” has read the label instead of counting the stack — the single most common line-plot error, and worth catching on problem 1.

2. Find the tallest stack. Which number of books has the tallest stack?

Compare the heights: the stack above 2 has four X's, above 3 has six, above 4 has three, above 5 has two, above 6 has one. The tallest is the one above 3. 3

That stack holds six X's, so six children read three books — more than any other number of books.

Watch the wording. “Which number of books” asks for a label from the bottom of the plot; “how many children” asks for the size of a stack. Both questions point at the same tall column, and the two answers are different numbers (3 and 6). Ask the child which one they wrote and why.

3. How many children in all?

Every child put exactly one X on the plot, so counting the children means counting every X. Take the stacks one at a time so nothing gets counted twice:

above 2: four above 3: six above 4: three above 5: two above 6: one

$4 + 6 + 3 + 2 + 1 = 16$ children in Room 4. 16

A check worth teaching. If the total comes out bigger than the class, a stack was counted twice; if it comes out smaller, a stack was missed. Crossing off each stack as it is added is what stops both.

4. Find the shortest stack. Which number of books was read by the fewest children?

The shortest stack is the one above 6: it holds a single X. So 6 books were read by the fewest children — exactly one child. 6

The trap is position. The stack above 2 is at the far left, and “fewest” feels like it should live at that end. It does not: left-to-right on a line plot is about the size of the *thing measured*, and up-and-down is about *how many*. Have the child point at the shortest column rather than the leftmost one.

5. How many more? How many more children read 3 books than 4 books?

Count the two stacks: above 3 there are six X's, above 4 there are three.

“How many more” is a subtraction: $6 - 3 = 3$. Three more children read 3 books than read 4 books. 3

If they answered 9: they added the stacks instead of comparing them. Adding tells you how many children are in the two stacks together; subtracting tells you how much taller one stack is than the other. Line them up with two fingers, one on each stack, and count the X's that stick out above the shorter one.

6. Finish the line plot. How many X's did you draw above 6?

Working through Ms. Ruiz's list and crossing off the lengths already drawn (5, 7 and 8), what is left is the four 6's and the single 10.

So four new X's go above the 6, and one goes above the 10. 4

The finished plot has two X's above 5, four above 6, three above 7, two above 8, none above 9, and one above 10.

How to check a plot a child has built. Count the X's on the plot and count the numbers on the list — both must come to twelve. If the plot has more, a length was plotted twice; if it has fewer, one was skipped. That check does not need the answer key.

7. How many shells?

Count the X's, not the numbers along the bottom: $2 + 4 + 3 + 2 + 0 + 1 = 12$ shells. 12

This matches the twelve lengths on Ms. Ruiz's list, which is exactly the check the plot is supposed to survive.

If they answered 6: they counted the marks on the number line — 5, 6, 7, 8, 9, 10 — instead of the X's. Those six numbers are the lengths that were *possible*, not the shells that were *found*. Ask how many shells are above the 9 (none), and the difference between the two counts usually lands.

8. Compare two stacks. How many more shells were 6 cm long than 5 cm long?

Above 6 there are four X's; above 5 there are two.

$4 - 2 = 2$, so two more shells were 6 cm long than were 5 cm long. 2

If they answered 1: they subtracted the two lengths ($6 - 5$) instead of the two stack sizes. The question asks about shells, so both numbers must be counts of X's. This is the same swap as problem 1, one step further on, and it is worth naming out loud every time it appears.

9. Longest and shortest. How much longer was the longest shell than the shortest one?

The furthest-right X sits above 10, so the longest shell was 10 cm. The furthest-left X sits above 5, so the shortest was 5 cm.

$10 - 5 = 5$, so the longest shell was five centimetres longer than the shortest. 5 cm

This answer is a length, and the unit matters. Problems 7 and 8 answered in shells; this one answers in centimetres, because the two numbers being subtracted came from the bottom of the plot rather than from the stacks.

If they answered 6: they counted the marks from 5 to 10 and included both ends. Counting marks answers “how many different lengths could there be”, not “how much longer”. Six marks, five steps between them — walking a finger along the line while counting the *jumps* makes the difference visible.

10. Explain it to Sam.

(a) There are no X's above the 9, so the number of shells that were 9 cm long is 0

(b) (Open explanation — listen to the reason.)

A model answer. “The empty space above the 9 does not mean Ms. Ruiz forgot it. It means she measured shells and none of them turned out to be 9 cm long. Zero is a real answer. The 9 has to stay on the line because the numbers have to go up one at a time and stay evenly spaced — if you rubbed the 9 out, the 10 would sit right next to the 8 and it would look as though the longest shell was only one centimetre longer than an 8 cm one.”

Marking guide. Full credit for any answer that says the gap means *no shells were that long*, or that the numbers must stay in order and evenly spaced so the gaps are readable. Partial credit for “Sam is wrong, there were no 9 cm shells” with nothing said about why the 9 stays. Not sufficient: “because it is on the line” — true, but it repeats the plot instead of explaining it.

If the child agrees with Sam: cover the 9 with a finger and ask them to read the plot again. Most children immediately say the last shell now looks 9 cm long, which is the whole argument.

Open explanation — see model answer.